

ERGODIC THEORY, TORUN, OCTOBER 2014 EXTRA EXERCISES.

JON AARONSON

§1 EXISTENCE AND UNIQUENESS OF acims

¶1.1 Here you construct an absolutely continuous, T -invariant measure from an invariant measure for an induced transformation by Kac's formula.

Let T be a nonsingular transformation of $(X, \mathcal{B}, m)$, and suppose that the return time function to $A \in \mathcal{B}_+$ given by $\varphi_A(x) := \min \{n \geq 1 : T^n x \in A\} \leq \infty$ (where $\min \emptyset := \infty$) is finite m -a.e. on A .

Suppose that $q \ll m|_A$ is a σ -finite, T_A -invariant measure. Set, for $B \in \mathcal{B}$,

$$\mu(B) := \int_A \sum_{k=0}^{\varphi_A-1} 1_B \circ T^k dq.$$

Show that

- 0) $\mu(B) = \sum_{k=0}^{\infty} q(A \cap T^{-k}B \setminus \bigcup_{j=1}^k T^{-j}A)$;
- 1) $\mu \ll m$ is a σ -finite, T -invariant measure,
- 2) if $(A, \mathcal{B}(A), q, T_A)$ is conservative, then so is $(X, \mathcal{B}, \mu, T)$ and $X \stackrel{\mu}{=} \bigcup_{k=0}^{\infty} T^{-k}A$;
- 3) if $(A, \mathcal{B}(A), q, T_A)$ is conservative, ergodic, then so is $(X, \mathcal{B}, \mu, T)$.

¶1.2 Let T be a conservative, ergodic, nonsingular transformation of $(X, \mathcal{B}, m)$. Here, you show that up to multiplication by constants, there is at most one m -absolutely continuous, σ -finite T -invariant measure.

Note that this is easy(!) in case T is invertible. In general the following exercises provide a proof⁶:

- (i) Suppose that $\mu \ll m$ is a σ -finite T -invariant measure, then $\mu \sim m$.

Hint Let $h = \frac{d\mu}{dm}$, then $\widehat{T}h = h$ and $\mu(A) = \int_A h dm = 0$ iff $A \subset [h = 0] \bmod m$.

- (ii) Suppose that $m \in \mathcal{P}(X, \mathcal{B})$, $m \circ T^{-1} = m$ and that $\mu \ll m$ is another T -invariant probability, then $\mu \equiv m$.

(iii) Suppose that T is a conservative, ergodic, measure preserving transformation of $(X, \mathcal{B}, \mu)$, and $\nu \ll \mu$ is a σ -finite T -invariant measure, then $\nu = c\mu$ for some $c \in \mathbb{R}_+$.

Hint $\forall A \in \mathcal{B}, 0 < \mu(A) < \infty, \exists 0 < c < \infty$ s.t. $\nu(B) = c\mu(B) \quad \forall B \in \mathcal{B}(A) \dots$

¶1.3 Here you show by example that conservativity in ¶2 cannot be dropped.

Let $(\Omega \times \mathbb{Z}, \mathcal{B}(\Omega \times \mathbb{Z}), \mu_p \times \#, S_f)$ be the totally dissipative, exact MPT considered in Exercise 16 of the minicourse notes.

Define $N : \Omega \times \mathbb{Z} \rightarrow [0, \infty]$ by

$$N := \sum_{k=0}^{\infty} 1_{\Omega \times \{0\}} \circ S_f^k$$

and define $h : \Omega \times \mathbb{Z} \rightarrow [0, \infty)$ by

$$h(x, z) = \begin{cases} 0 & N(x, z) > 1, \\ 1 & N(x, z) = 1, \quad z < 0, \\ \frac{1}{2} & \text{else.} \end{cases}$$

(v) Show that h is the density of a σ -finite, m -absolutely continuous, S_f -invariant measure which is not a constant multiple of m .¹

§2 MARKOV SHIFTS

Let S be a countable set, $T : S^{\mathbb{N}} \rightarrow S^{\mathbb{N}}$ be the *shift* map, defined by $T(x_1, x_2, \dots) = (x_2, x_3, \dots)$.

¶2.1 Markov measure. Given a *stochastic matrix* P on S ($p : S \times S \rightarrow \mathbb{R}, p_{s,t} \geq 0, \sum_{t \in S} p_{s,t} = 1 \quad \forall s \in S$) and a probability π on S such that $\pi_s > 0 \quad \forall s \in S$, show existence of a probability $m = m_\pi \in \mathcal{P}(S^{\mathbb{N}})$ such that

$$m([s_1, \dots, s_n]) = \pi_{s_1} p_{s_1, s_2} \cdots p_{s_{n-1}, s_n} \quad \forall s_1, \dots, s_n \in S^n, \quad n \in \mathbb{N}$$

where $[s_1, \dots, s_n] := \{x = (x_1, x_2, \dots) \in S^{\mathbb{N}} : x_1 = s_1, \dots, x_n = s_n\}$.

The measure m_π is aka *Markov measure*.

The *Markov shift* of P (with initial distribution π) is the quadruple $(S^{\mathbb{N}}, \mathcal{B}, m_\pi, T)$ where $\mathcal{B} = \sigma(\{[s_1, \dots, s_n] : s_1, \dots, s_n \in S^n, \quad n \in \mathbb{N}\})$ and T is the shift.

¹For the general result, see Aaronson J., Meyerovitch, T., Colloq. Math. **110**, (2008), 193–199.

¶2.2 Non-singularity.

Show that T

(i) is a null preserving transformation of $(S^{\mathbb{N}}, \mathcal{B}, m)$ with

$$\frac{dm \circ T^{-1}}{dm}(x) = \sum_{s \in S} \frac{\pi_s p_{s,x_1}}{\pi_{x_1}}$$

with transfer operator given by

$$\widehat{T}^k 1_{[t]}(x) = \frac{\pi_t}{\pi_{x_1}} p_{t,x_1}^{(k)} \quad \forall k \in \mathbb{N}, t \in S$$

where $p_{s,t}^{(1)} := p_{s,t}$ and $p_{s,t}^{(k+1)} := \sum_{u \in S} p_{s,u}^{(k)} p_{u,t}$;
and T

(ii) is nonsingular iff

$$\forall t \in S, \exists s \in S, p_{s,t} > 0.$$

¶2.3 Hopf decomposition.

A state $s \in S$ is called *recurrent* if $\sum_{n=0}^{\infty} p_{s,s}^{(n)} = \infty$ and *transient* otherwise.

Show that $\mathfrak{C}(T) = \bigcup_{s \in S_r} [s] \mod m$ where S_r denotes the collection of recurrent states.

¶2.4 Irreducibility and conservativity.

A stochastic matrix is called *irreducible* if

$$\forall s, t \in S, \exists n \in \mathbb{N} \ni p_{s,t}^{(n)} > 0.$$

Show that if the stochastic matrix P is irreducible and at least one state is recurrent, then all states are recurrent and the Markov shift $(S^{\mathbb{N}}, \mathcal{B}, m_\pi, T)$ is conservative.

The next exercises complete a proof of the

Harris-Robbins Theorem *If the stochastic matrix P is irreducible and recurrent (i.e. all states are recurrent), then the Markov shift $(S^{\mathbb{N}}, \mathcal{B}, m_\pi, T)$ is conservative, ergodic and $\exists \mu \sim m_\pi$ σ -finite and T -invariant.*

By the ¶2.4, $(S^{\mathbb{N}}, \mathcal{B}, m_\pi, T)$ is conservative.

¶2.5 Exactness of induced transformation.

Fix $s \in S$, and let m_s be that constant multiple of $m|_{[s]}$ with $m_s([s]) = 1$. Show that then $T_{[s]} ([s], \mathcal{B} \cap [s], m_s, T_{[s]})$ is an exact, probability preserving transformation.

Hint:

Let $\alpha = \{[s, t_1, \dots, t_{n-1}, s] : n \geq 1, t_1, \dots, t_{n-1} \neq s\}$, then α is a partition mod m_s of $[s]$, $\sigma(\bigcup_{n \geq 0} T_{[s]}^{-n} \alpha) = \mathcal{B} \cap [s]$ and $\{T_{[s]}^{-n} \alpha : n \geq 0\}$ are statistically independent wrt $m_s, \dots$.

¶2.6 Ergodicity.

Show that $(S^{\mathbb{N}}, \mathcal{B}, m_\pi, T)$ is ergodic.

¶2.7 Markov invariant measure.

For $s \in S$, let

$$\mu_s(B) = \int_{[s]} \left(\sum_{k=0}^{\varphi_{[s]}^{-1}-1} 1_B \circ T^k \right) dm_s.$$

Show that

- (i) $\mu_s \circ T^{-1} = \mu_s \sim m \quad \forall s \in S;$
- (ii) $\mu_t \equiv \mu_s([t])\mu_s$ and $0 < \mu_s([t]) < \infty$ for $s, t \in S$;
- (iii) $\mu_s([t_1, \dots, t_n]) = \mu_s([t_1]) \prod_{k=1}^{n-1} p_{t_k, t_{k+1}};$
- (iv) $\sum_{t \in S} \mu_s([t]) p_{t,u} = \mu_s(T^{-1}[u]) = \mu_s([u]).$

¶2.8 Pointwise dual ergodicity.

Suppose that $P : S \times S \rightarrow [0, 1]$ is irreducible, recurrent. Consider now the measure preserving transformation $(X, \mathcal{B}, \mu, T)$ where $\mu = \mu_s$ as in ¶9.5. Show that

$$(i) \quad \frac{1}{a_n(T)} \sum_{k=0}^{n-1} \widehat{T}^k f \xrightarrow{n \rightarrow \infty} \int_X f d\mu \text{ a.e.} \quad \forall f \in L^1(\mu)$$

where $a_n(T) \sim \sum_{k=0}^{n-1} p_{s,s}^{(k)}$ and deduce Doeblin's *ratio limit theorem*:

$$(ii) \quad \frac{1}{a_n(T)} \sum_{k=0}^{n-1} p_{u,v}^{(k)} \xrightarrow{n \rightarrow \infty} m_v \quad \forall u, v \in S$$

where $m_t := \mu_s([t]).$